AI tools and prompt engineering

AI training workshop

Rahul Dave

Writing and document questions

Turn rough notes into a draft.

Extract facts with supporting passages.

Tools for different kinds of work

Work	Feature	Example result
Writing and revision	Chat	A draft from rough notes
Questions about a document	File input or document search	A fact and its source passage
Current information	Web search	A summary with dated sources

ChatGPT and Gemini are examples. Features depend on the account and workspace.

A prompt is a request

Describe the task and the result you need.

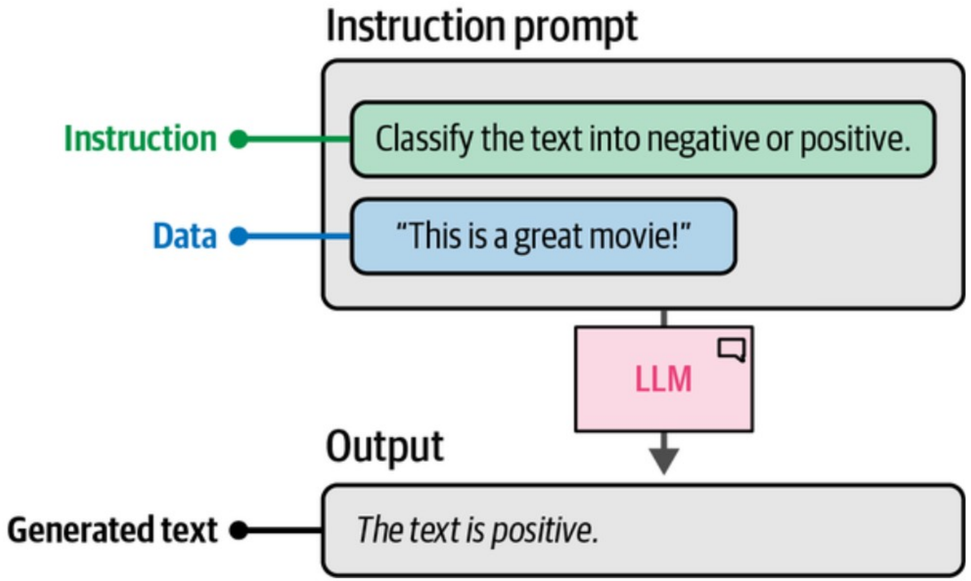
Turn these notes into a brief update for our team.

A short prompt can be enough for a simple task.

The parts of a useful prompt

Task	The work the model should do
Context	The information it can use
Output	The form the answer should take
Checks	The conditions the answer must meet

Instructions and output format



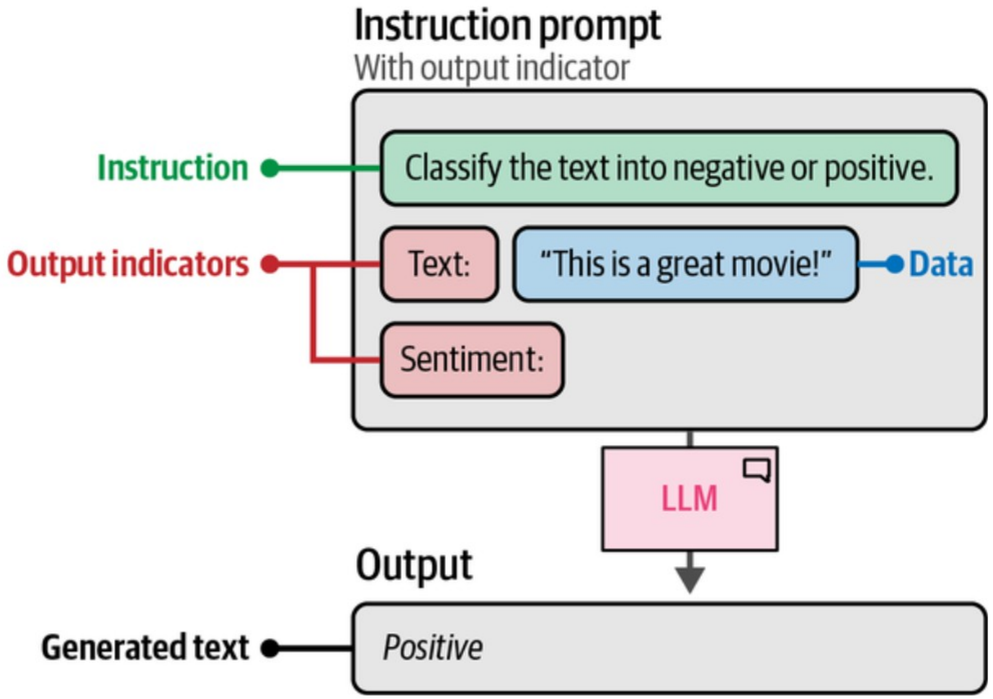


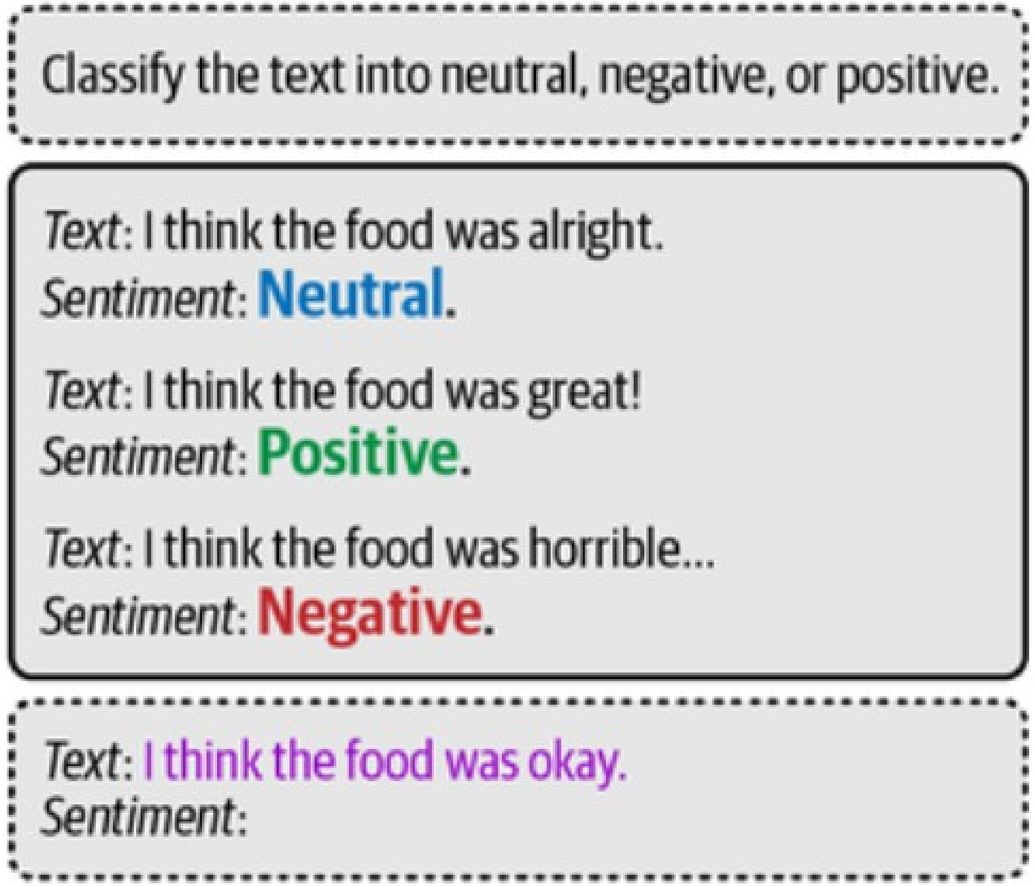
Diagram: Jay Alammar and Maarten Grootendorst, from the original teaching deck

A prompt with useful details

Task	Rewrite these notes for the project team.
Context	The launch moved to 12 May. Priya owns the draft.
Output	Write three short bullets.
Checks	Keep the date and owner unchanged.

Illustrative example

Examples show the pattern



Examples pair a review with a label.

The final review has no label.

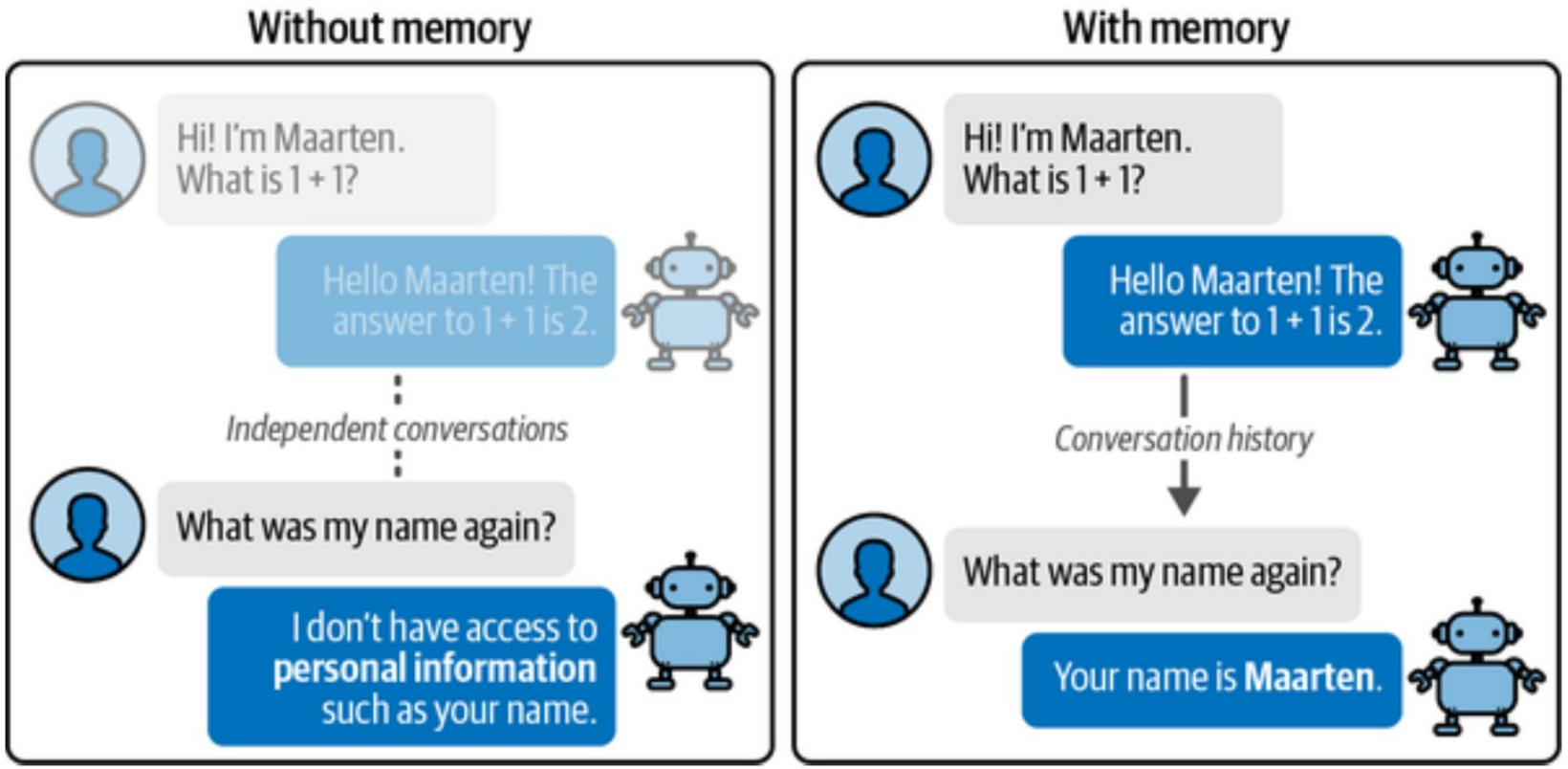
Diagram: Jay Alammar and Maarten Grootendorst, from the original teaching deck

An example for missing information

Input	Requested course date
The course starts on 12 May.	12 May
Registration opens soon.	Not stated

Examples can show how to handle a missing fact.

Earlier messages can affect the answer



Conversation history supplies context for later replies.

Diagram: Jay Alammar and Maarten Grootendorst, from the original teaching deck

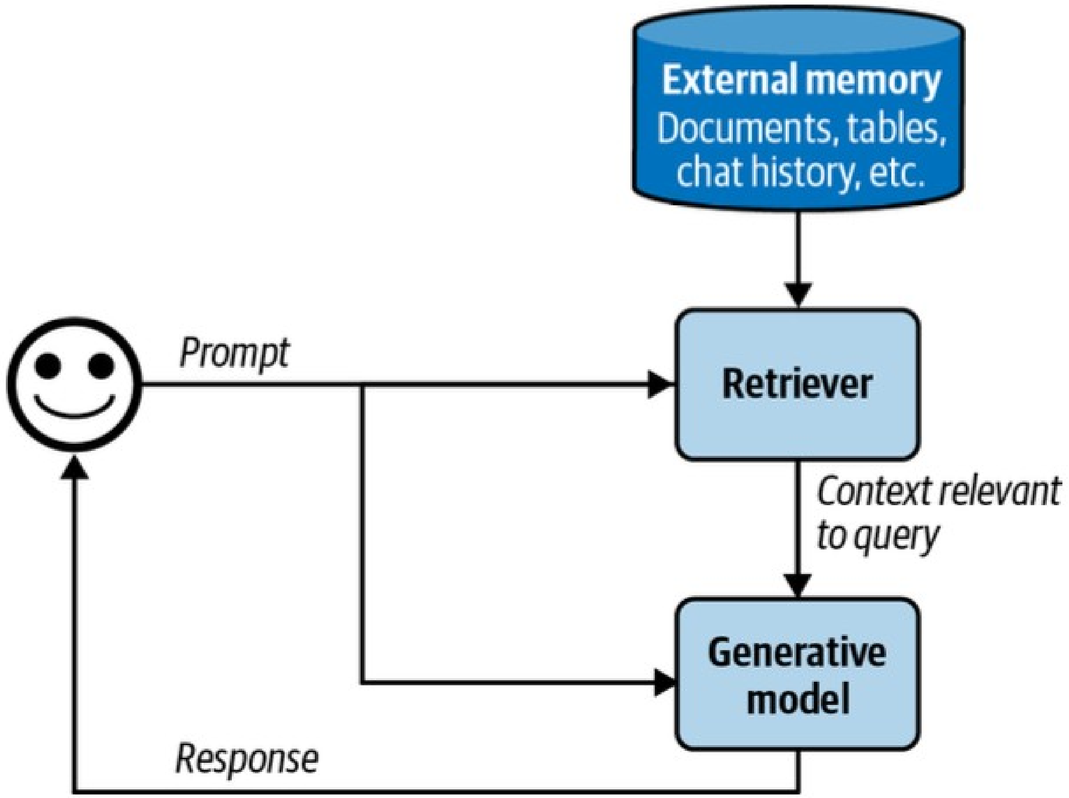
Credcore's document task

Extract a requested fact from a contract.

Keep the passage that supports the answer.

A person checks the result against the document.

Relevant passages become context



Retrieval finds relevant passages.

The model uses them to write an answer.

Retrieval augmented generation is often called RAG.

Diagram: Chip Huyen, AI Engineering, reproduced in the Credcore notes

Practice document

Page 1. Booking terms

The course starts on 12 May.
The standard fee is $600 per participant.
The fee for partner organizations is $450 per participant.

Page 2. Payment terms

Invoices are due within 20 calendar days of receipt.
Cancellation requires 10 calendar days' notice.

Fictional text for the workshop exercise

A broad summary request

Prompt

Summarize the booking terms.

Illustrative answer

Participants pay $600 and have 20 days to pay.

A document extraction prompt

Extract the standard fee, partner fee, payment deadline, and minimum group size.

Use only the supplied document.

Return a table with the field, answer, quote, and page.

If the document gives no answer, write "Not stated".

Keep the two fees separate.

Practice in pairs

Copy the practice text into your approved chat assistant.

Run a prompt of your own.

Check every answer against the document.

Change one instruction and run it again.

Eight minutes. Keep both prompts and both answers.

An answer with supporting passages

Field	Answer	Supporting quote	Page
Standard fee	$600 per participant	The standard fee is $600 per participant.	1
Partner fee	$450 per participant	The fee for partner organizations is $450 per participant.	1
Payment deadline	20 calendar days after receipt	Invoices are due within 20 calendar days of receipt.	2
Minimum group size	Not stated	No supporting passage	None

Worked answer for the fictional practice document

Checking an answer

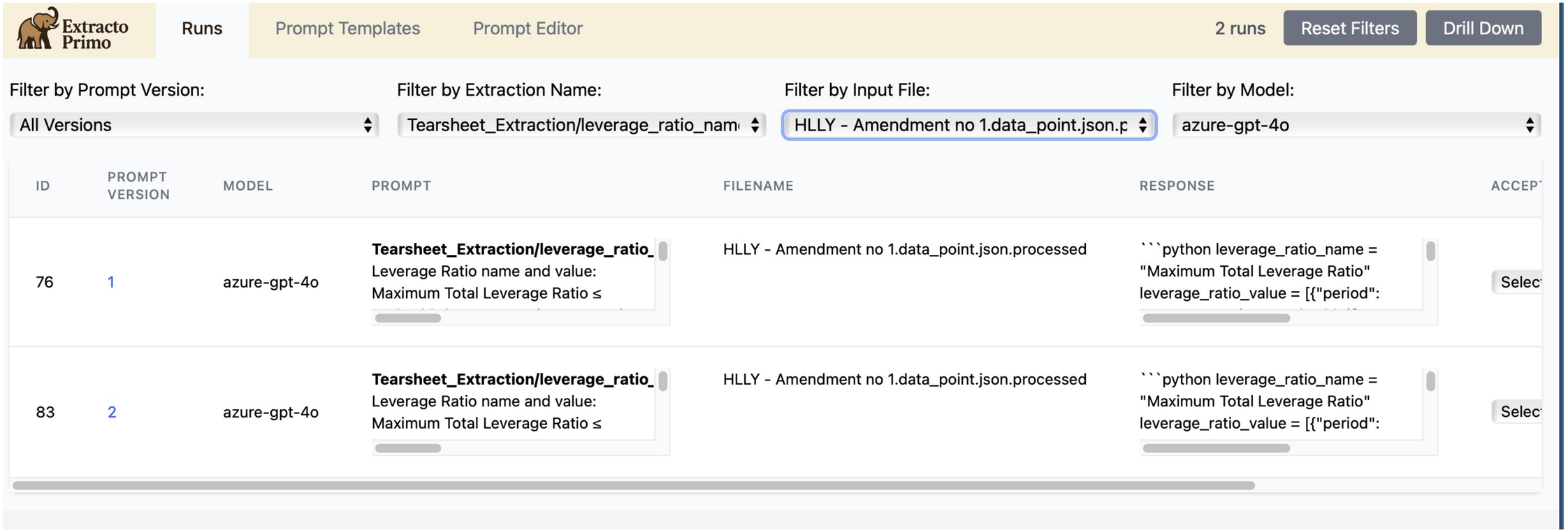
Task	Every requested field appears.
Evidence	Each quote supports its answer.
Exceptions	The two fees stay separate.
Missing facts	Unstated details stay "Not stated".

A revision for a missing condition

Observed error	The answer says "20 days".
Missing detail	The period starts when the invoice arrives.
Prompt revision	State when each payment period begins.

Test the revision on the same document. Compare the answers.

Prompt versions and results



The record connects each result to the prompt that produced it.

Screenshot from Rahul Dave's Credcore work, LLMAndPrompt.key, slide 38

AI can help check an answer

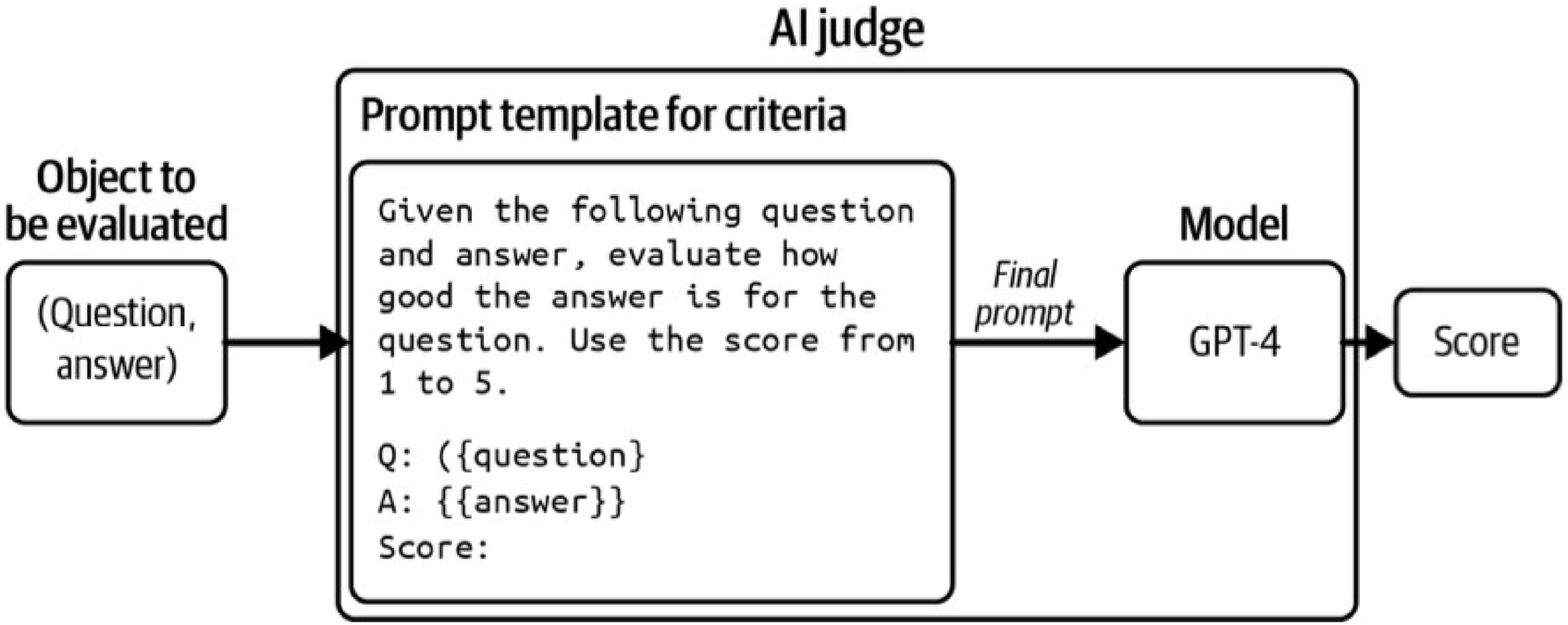
Compare the answer with the document.

For each field, report supported, unsupported, or missing.

Quote the passage behind each judgment.

Give the checker the document and the answer.

An AI judge



People compare the judge's feedback with their own reviews.

Diagram: Chip Huyen, AI Engineering, reproduced in the Credcore notes

Limits of an AI judge

A second model can repeat the first model's mistake.

Important answers need review against the original source.

A check must test the answer against clear criteria.

A prompt for your own task

Task	Describe one result you need.
Context	Supply the information needed for that result.
Output	Choose a form you can use.
Checks	State how you will judge the answer.

An example can show the pattern you want.

Next session: AI agents

An agent can use tools during a task.

We will follow a search across several documents.

The task and the checks still come from us.

Sources and further reading

Rahul Dave's teaching slides and Credcore notes

Alammar and Grootendorst, Hands-On Large Language Models

Chip Huyen, AI Engineering

Official prompting guidance